
MODEL SELECTION AND VALIDATION REPORT (only_teve_biomass model)

OBSERVATIONS:3382

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (only_teve_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	7.04	0.22
teve	-0.44	0.02
sbt_mean	0.37	0.06
sic_mean	0.06	0.05
sbt_sd	0.13	0.03
sst_sd	0.03	0.03
log_chla_mean	0.02	0.02
depth	0.07	0.03

Dispersion parameter: 0.75

Spatiotemporal AR1 correlation (rho): 0.91

Matérn range: 318.43

Spatiotemporal marginal AR1 SD: 0.88

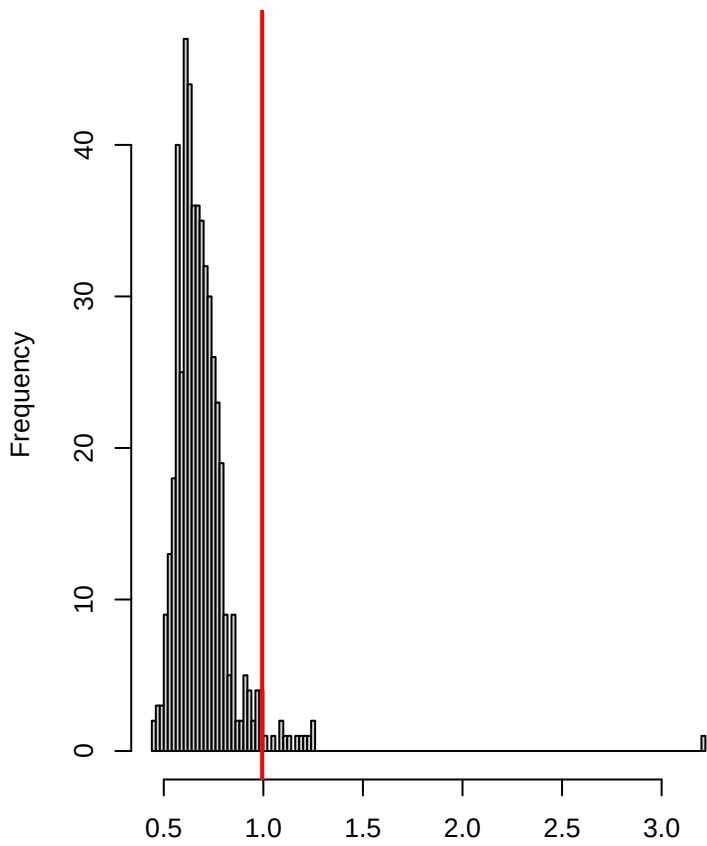
REML criterion at convergence: 27028.775

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

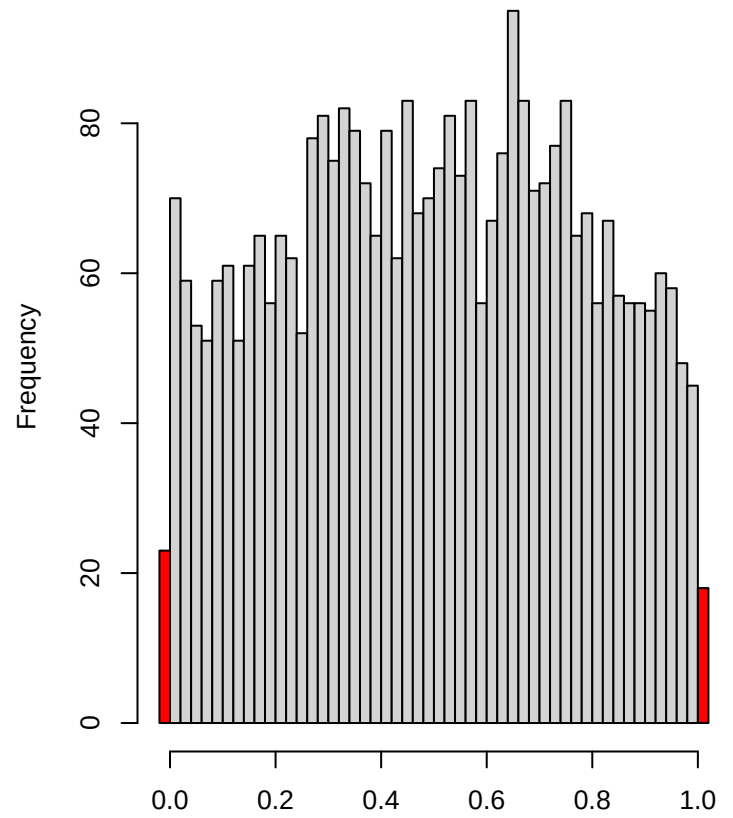
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	12	54081.55	0.02398124
spatiotemporal_random_fields_ar1_with_anisotropy	14	54081.57	273.61367063
spatiotemporal_random_fields_iid_with_anisotropy	13	54355.19	4.64020993
spatiotemporal_random_fields_iid	11	54359.83	263.07256272
spatial_random_fields	11	54622.90	2.19810994
spatial_random_fields_with_anisotropy	13	54625.10	583.26215568
error_only	9	55208.36	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



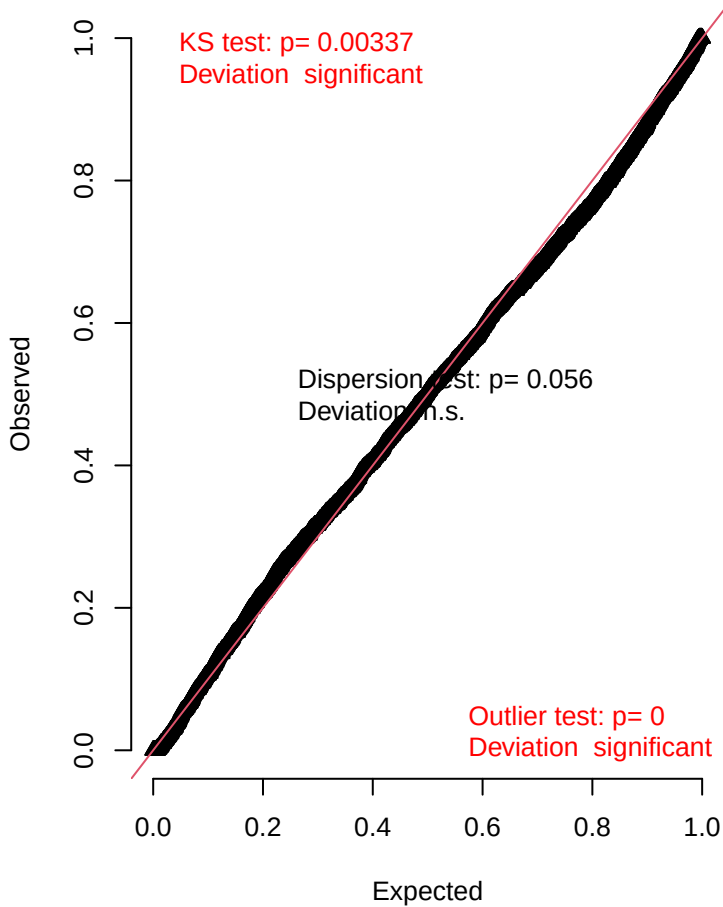
Simulated values, red line = fitted model. p-value (two.sided) =

Outlier test significant

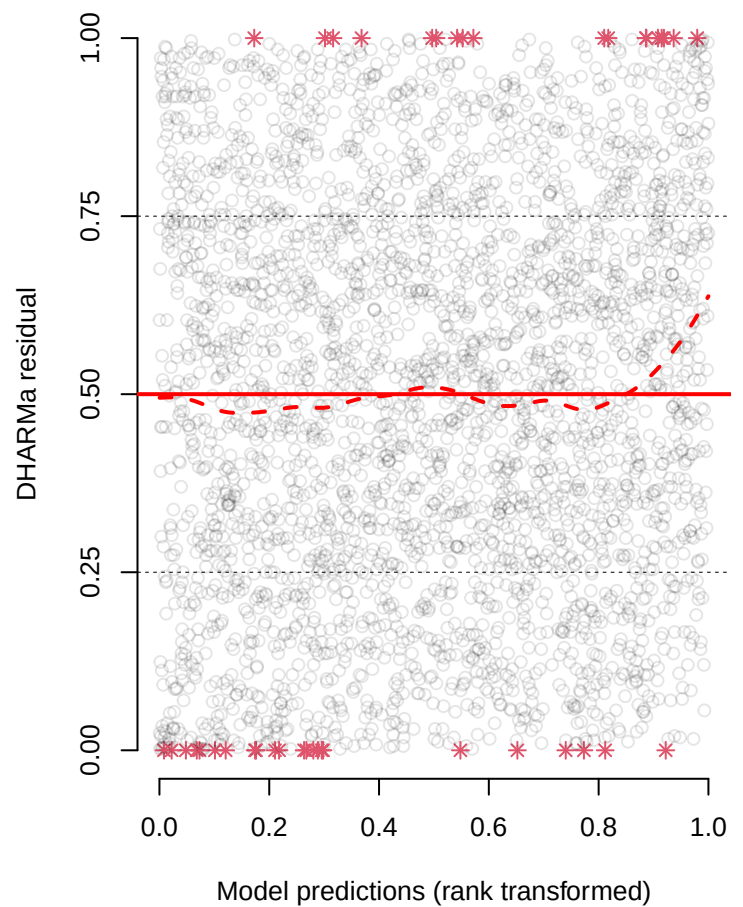


Residuals (outliers are marked red)

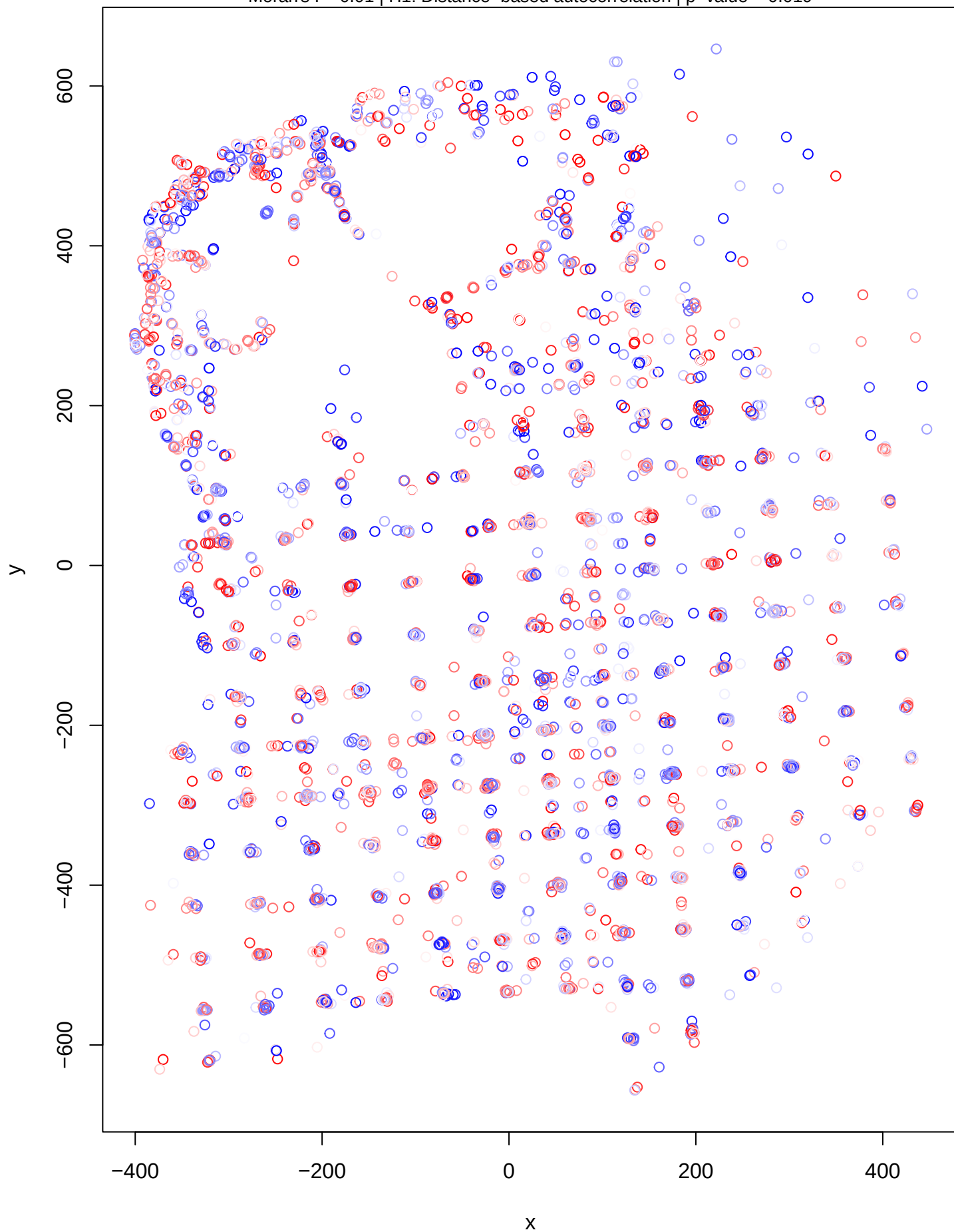
QQ plot residuals



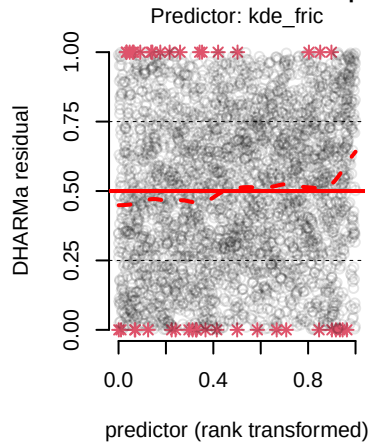
DHARMa residual vs. predicted



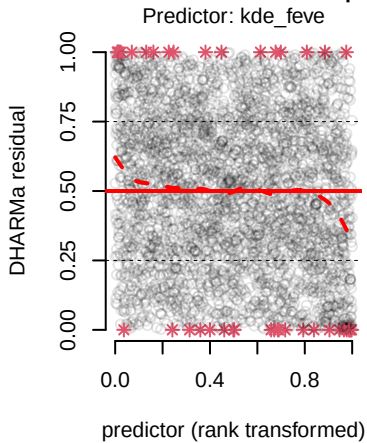
DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.01 | H1: Distance-based autocorrelation | p-value = 0.019



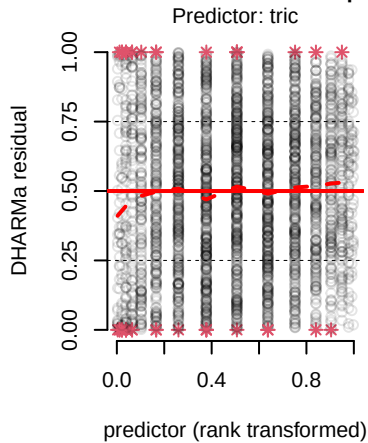
DHARMa residual Residual vs. pred



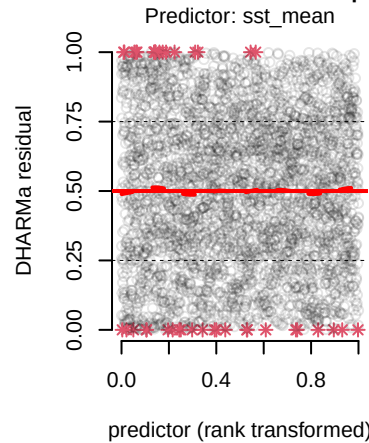
DHARMa residual Residual vs. pred



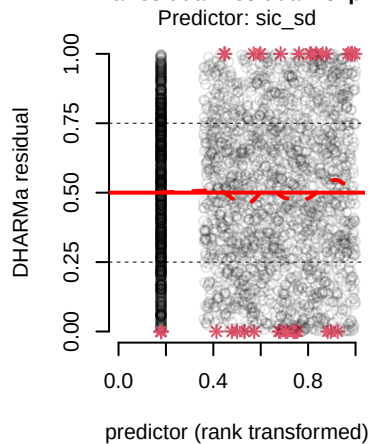
DHARMa residual Residual vs. pred



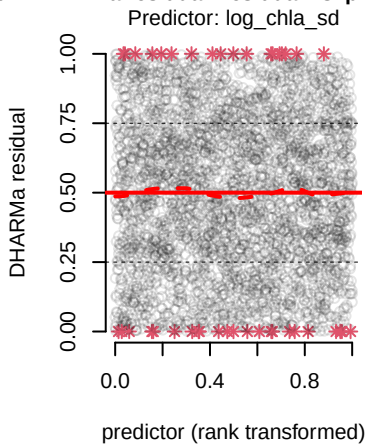
DHARMa residual Residual vs. pred



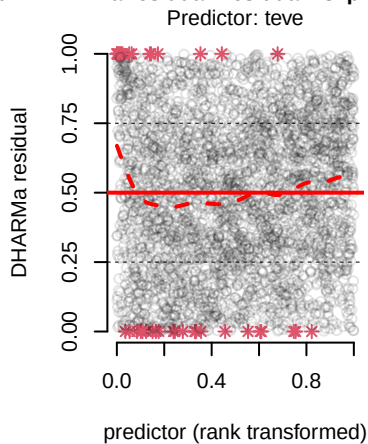
DHARMa residual Residual vs. pred



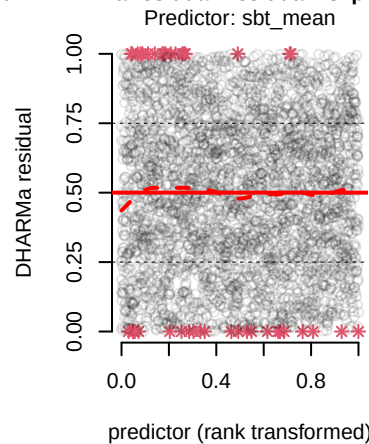
DHARMa residual Residual vs. pred



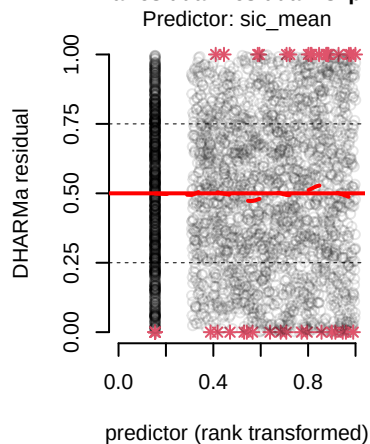
DHARMa residual Residual vs. pred



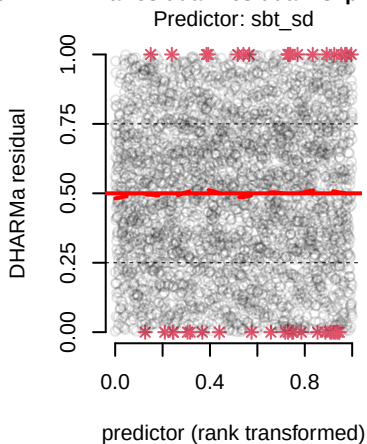
DHARMa residual Residual vs. pred



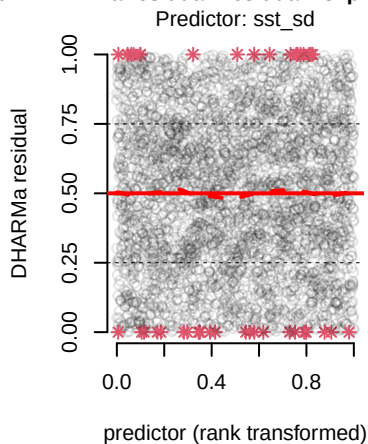
DHARMa residual Residual vs. pred



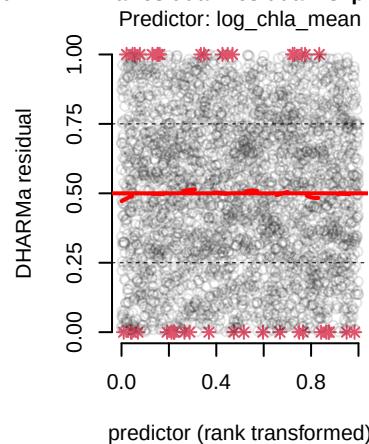
DHARMa residual Residual vs. pred



DHARMa residual Residual vs. pred



DHARMa residual Residual vs. pred



DHARMa residual Residual vs. pred

